

- If the patient has experienced an episode of complex sleep behaviors after taking a sleep medication
- If there is a proven allergy to zolpidem
- For Intermezzo, if the patient has fewer than 4 hours of bedtime remaining before the planned time of waking

Special Populations

Renal Impairment

- No dose adjustment necessary
- Patients should be monitored

Hepatic Impairment

- Recommended dose 5 mg (immediate-release), 6.25 mg (controlled-release), 1.75 mg (Intermezzo)
- Patients should be monitored

Cardiac Impairment

- No available data

Elderly

- Recommended initial dose: 5 mg (immediate-release), 6.25 mg (controlled-release), 1.75 mg (Intermezzo)

- Elderly may have increased risk for falls, confusion



Children and Adolescents

- Safety and efficacy have not been established
- Long-term effects of zolpidem or zolpidem CR in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored
- Hallucinations in children ages 6–17 have been reported



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In animal studies, oral administration of zolpidem did not indicate a risk for adverse effects on fetal development at clinically relevant doses
- Infants whose mothers took sedative hypnotics during pregnancy may experience some withdrawal symptoms

- Neonatal flaccidity has been reported in infants whose mothers took sedative hypnotics during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed

The Art Of Psychopharmacology

Potential Advantages

- Patients who require long-term treatment, especially CR formulation

Potential Disadvantages

- More expensive than some other sedative hypnotics

Primary Target Symptoms

- Time to sleep onset
- Total sleep time
- Nighttime awakenings

Pearls

- * One of the most popular sedative hypnotic agents in psychopharmacology

- Zolpidem has been shown to increase the total time asleep and to reduce the amount of nighttime awakenings
- Zolpidem CR may be even more effective on these sleep parameters than immediate-release zolpidem due to prolonged drug delivery
- * May be preferred over benzodiazepines because of its rapid onset of action, short duration of effect, and safety profile
- In some patients, zolpidem blood levels may be high enough the morning after use to impair activities that require alertness, including driving; this has prompted the US FDA to issue new dosing requirements
- Specifically, because clearance of zolpidem is slightly slower in women than in men, the FDA has required that the recommended dose be lowered for women
- The FDA also recommends that healthcare professionals consider prescribing lower doses for men as well
- A low-dose zolpidem product is approved for middle-of-the-night awakenings by sublingual administration
- May not be ideal for patients who desire immediate hypnotic onset and eat just prior to bedtime
- Not a benzodiazepine itself, but binds to benzodiazepine receptors
- May have fewer carryover side effects than some other sedative hypnotics

- May cause less dependence than some other sedative hypnotics, especially in those without a history of substance abuse

Suggested Reading

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J Am Acad Psychiatry Law 2011 ;**39** (4):535–42 .

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benzodiazepines: a review . *Pharmacol Biochem Behav* 1998 ;**61** :253–69 .

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Soyka M , Bottlender R , Moller HJ . Epidemiological evidence for a low
abuse potential of zolpidem . *Pharmacopsychiatry* 2000 ;**33** :138–41 .

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Toner LC , Tsambiras BM , Catalano G , Catalano MC , Cooper DS .
Central nervous system side effects associated with zolpidem treatment .
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Zonisamide

Therapeutics

Brands

- Zonegran
- Excegran
- *see index for additional brand names*

Generic?

Yes

Class

- Anticonvulsant, voltage-sensitive sodium channel modulator; T-type calcium channel modulator; structurally a sulfonamide

Commonly Prescribed for

(bold for FDA approved)

- **Adjunct therapy for partial seizures in adults with epilepsy**
- Bipolar disorder
- Chronic neuropathic pain
- Migraine
- Parkinson's disease

- Psychotropic drug-induced weight gain
- Binge eating disorder

How the Drug Works

- Unknown
- Modulates voltage-sensitive sodium channels by an unknown mechanism
- Also modulates T-type calcium channels
- Facilitates dopamine and serotonin release
- Inhibits carbonic anhydrase

How Long Until It Works

- Should reduce seizures by 2 weeks
- Onset of action as well as convincing therapeutic efficacy have not been demonstrated for uses other than adjunctive treatment of partial seizures

If It Works

- The goal of treatment is complete remission of symptoms (e.g., seizures, pain, mania, migraine)
- Would currently only be expected to work in a subset of patients for conditions other than epilepsy as an adjunctive treatment to agents with better demonstration of efficacy

If It Doesn't Work (for Conditions Other Than Epilepsy)

- May be effective only in patients who fail to respond to agents with proven efficacy, or it may not work at all
- Consider increasing dose or switching to another agent with better demonstrated efficacy



Best Augmenting Combos for Partial Response or Treatment Resistance

- Zonisamide is itself a second-line augmenting agent to numerous other agents in treating conditions other than epilepsy, such as bipolar disorder, chronic neuropathic pain, and migraine

Tests

- Consider baseline and periodic monitoring of renal function

Side Effects

How Drug Causes Side Effects

- CNS side effects theoretically due to excessive actions at voltage-sensitive ion channels
- Weak inhibition of carbonic anhydrase may lead to kidney stones
- Serious rash theoretically an allergic reaction

Notable Side Effects

* Sedation, depression, difficulty concentrating, agitation, irritability, psychomotor slowing, dizziness, ataxia

- Headache
- Nausea, anorexia, abdominal pain, vomiting
- Kidney stones
- Elevated serum creatinine and blood urea nitrogen



Life-Threatening or Dangerous Side Effects

- Rare serious rash (Stevens-Johnson syndrome, toxic epidermal necrolysis) (sulfonamide)
- Rare oligohidrosis and hyperthermia (pediatric patients)
- Rare blood dyscrasias (aplastic anemia; agranulocytosis)
- Sudden hepatic necrosis
- Sudden unexplained deaths have occurred (unknown if related to zonisamide use)
- Rare activation of suicidal ideation and behavior (suicidality)

Weight Gain



- Reported but not expected
- * Patients may experience weight loss

Sedation



- Many experience and/or can be significant in amount
- Dose-related
- Can wear off with time but may not wear off at high doses

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take more of the dose at night to reduce daytime sedation
- Lower the dose
- Switch to another agent

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- 100–600 mg/day in 1–2 doses

Dosage Forms

- Capsule 25 mg, 50 mg, 100 mg

How to Dose

- Initial 100 mg/day; after 2 weeks can increase to 200 mg/day; dose can be increased by 100 mg/day every 2 weeks if necessary and tolerated; maximum dose generally 600 mg/day; maintain stable dose for at least 2 weeks before increasing dose



Dosing Tips

- * Most clinical experience is at doses up to 400 mg/day
- No evidence from controlled trials of increasing response over 400 mg/day
- However, some patients may tolerate and respond to doses up to 600 mg/day
- Little experience with doses greater than 600 mg/day
- Side effects may increase notably at doses greater than 300 mg/day
- For intolerable sedation, can give most of the dose at night and less during the day

Overdose

- Can cause bradycardia, hypotension, respiratory depression

Long-Term Use

- Safe
- Consider periodic monitoring of blood urea nitrogen and creatinine

Habit Forming

- No

How to Stop

- Taper
- Epilepsy patients may seize upon withdrawal, especially if withdrawal is abrupt
- Rapid discontinuation may increase the risk of relapse in bipolar patients
- Discontinuation symptoms uncommon

Pharmacokinetics

- Plasma elimination half-life approximately 63 hours
- Metabolized in part by CYP450 3A4
- Partially eliminated renally



Drug Interactions

- Agents that inhibit CYP450 3A4 (such as nefazodone, fluvoxamine, and fluoxetine) may decrease the clearance of zonisamide, and increase plasma zonisamide levels, possibly requiring lower doses of zonisamide
- Agents that induce CYP450 3A4 (such as carbamazepine) may increase the clearance of zonisamide and decrease plasma

zonisamide levels, possibly requiring higher doses of zonisamide

- Enzyme-inducing antiepileptic drugs (carbamazepine, phenytoin, phenobarbital, and primidone) may decrease plasma levels of zonisamide
- Theoretically, zonisamide may interact with carbonic anhydrase inhibitors to increase the risk of kidney stones

Other Warnings/Precautions

- Depressive effects may be increased by other CNS depressants (alcohol, MAOIs, other anticonvulsants, etc.)
- Use with caution when combining with other drugs that predispose patients to heat-related disorders, including carbonic anhydrase inhibitors and anticholinergics
- * Life-threatening rashes have developed in association with zonisamide use; zonisamide should generally be discontinued at the first sign of serious rash
- Patient should be instructed to report any symptoms of hypersensitivity immediately (fever; flu-like symptoms; rash; blisters on skin or in eyes, mouth, ears, nose, or genital areas; swelling of eyelids, conjunctivitis, lymphadenopathy)
- Patients should be monitored for signs of unusual bleeding or bruising, mouth sores, infections, fever, and sore throat, as there may be an increased risk of aplastic anemia and agranulocytosis with zonisamide

- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

Do Not Use

- If there is a proven allergy to zonisamide or sulfonamides

Special Populations

Renal Impairment

- Zonisamide is primarily renally excreted
- Use with caution
- May require slower titration

Hepatic Impairment

- Use with caution
- May require slower titration

Cardiac Impairment

- No specific recommendations

Elderly

- Some patients may tolerate lower doses better

- Elderly patients may be more susceptible to adverse effects



Children and Adolescents

- Cases of oligohydrosis and hyperthermia have been reported
- Not approved for use in children under age 16
- Use in children for the expert only, with close monitoring, after other options have failed



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, www.aedpregnancyregistry.org
- Taper drug if discontinuing
- Seizures, even mild seizures, may cause harm to the embryo/fetus

- Lack of convincing efficacy for treatment of conditions other than epilepsy suggests risk/benefit ratio is in favor of discontinuing zonisamide during pregnancy for these indications

Breast Feeding

- Unknown if zonisamide is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued

The Art Of Psychopharmacology

Potential Advantages

- Treatment-resistant conditions
- Patients who wish to avoid weight gain

Potential Disadvantages

- Poor documentation of efficacy for off-label uses
- Patients noncompliant with twice daily dosing

Primary Target Symptoms

- Seizures
- Numerous other symptoms for off-label uses
- Patients with a history of kidney stones

Pearls

- Well studied in epilepsy
- * Much off-label use is based upon theoretical considerations rather than clinical experience or compelling efficacy studies
- Early studies suggest efficacy in binge eating disorder
- Early studies suggest possible efficacy in migraine
- Early studies suggest possible utility in Parkinson's disease
- Early studies suggest possible utility in neuropathic pain
- Early studies suggest some therapeutic potential for mood-stabilizing
- Chronic intake of caffeine may lower brain zonisamide concentrations and attenuate its anticonvulsant effects (based on animal studies)
- * Due to reported weight loss in some patients in trials with epilepsy, some patients with psychotropic-induced weight gain are treated with zonisamide
- Utility for this indication is not clear nor has it been systematically studied

- Phase 2 trials for the combination of zonisamide and bupropion as a treatment for obesity have been completed

Suggested Reading

Chadwick DW , Marson AG . Zonisamide add-on for drug-resistant partial epilepsy . *Cochrane Database Syst Rev* 2002 ;(2):CD001416 .

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Zopiclone

Therapeutics

Brands

- Imovane
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Non-benzodiazepine hypnotic; alpha 1 isoform selective agonist of GABA-A/benzodiazepine receptors

Commonly Prescribed for

(bold for FDA approved)

- Short-term treatment of insomnia



How the Drug Works

- May bind selectively to a subtype of the benzodiazepine receptor, the alpha 1 isoform

- May enhance GABA inhibitory actions that provide sedative hypnotic effects more selectively than other actions of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- * Long-term adaptations of zopiclone, a mixture of an active S enantiomer and an inactive R enantiomer, have not been well studied, but chronic studies of the active isomer eszopiclone suggest lack of notable tolerance or dependence developing over time

Notable Side Effects

- * Sedation
- * Dizziness, ataxia
- * Dose-dependent amnesia

* Hyperexcitability, nervousness

- Dry mouth, loss of appetite, constipation, bitter taste
- Impaired vision



Life-Threatening or Dangerous Side Effects

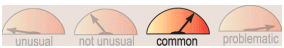
- Respiratory depression, especially when taken with other CNS depressants in overdose
- Rare angioedema

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- To avoid problems with memory, only take zopiclone if planning to have a full night's sleep
- Lower the dose
- Switch to a shorter-acting sedative hypnotic

- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- 7.5 mg at bedtime

Dosage Forms

- Tablet 5 mg, 7.5 mg scored

How to Dose

- No titration, take dose at bedtime



Dosing Tips

- Zopiclone should generally not be prescribed in quantities greater than a 1-month supply
- Risk of dependence may increase with dose and duration of treatment
- However, chronic treatment with alpha 1 selective non-benzodiazepine hypnotics may cause less tolerance or dependence than benzodiazepine hypnotics

Overdose

- Can be fatal; clumsiness, mood changes, sedation, weakness, breathing trouble, unconsciousness

Long-Term Use

- Not generally intended for use past 4 weeks

Habit Forming

- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

How to Stop

- Rebound insomnia may occur the first night after stopping
- If taken for more than a few weeks, taper to reduce chances of withdrawal effects

Pharmacokinetics

- Metabolized by CYP450 3A4
- Terminal elimination half-life approximately 3.5–6.5 hours



Drug Interactions

- Increased depressive effects when taken with other CNS depressants

- Theoretically, inhibitors of CYP450 3A4, such as nefazodone and fluvoxamine, could increase plasma levels of zopiclone

Other Warnings/Precautions

- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea
- Zopiclone should be administered only at bedtime
- Rare angioedema has occurred with sedative hypnotic use and could potentially cause fatal airway obstruction if it involves the throat, glottis, or larynx; thus if angioedema occurs treatment should be discontinued
- Sleep driving and other complex behaviors, such as eating and preparing food and making phone calls, have been reported in patients taking sedative hypnotics

Do Not Use

- If patient has myasthenia gravis